OIL PALM RESEARCH STRATEGIC PLAN 2023/24 - 2033/34

Contributing towards a self-sustainable vegetable oil industry in Uganda

NARO

FOR GOD AND MY COUNTRY

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ACRONYMS

NaCRRI	National Crops Resources Research Institute
OPR	Oil Palm Research
OPUL	Oil Palm Uganda Limited
BIDCO	Business and Industrial Development Corporations
ASARECA	Association for Strengthening Agricultural Research in Eastern and Central Africa
TICs	Technology Impact Centres
IMS	Information Management Systems
CBO	Community Based Organisation
TIMPs	Technologies, Innovations and Management Practices
EAC	East African Community
IR	Intermediate Result
KOPGT	Kalangala Oil Palm Growers Trust
MAAIF	Ministry of Agriculture Animal Industry and Fisheries
MOU	Memorandum of Understanding
NAADS	National Agricultural Advisory Services
NARO	National Agricultural Research Organisation
NARS	National Agricultural Research System
PESTEL	Political, Economic, Social, Technological, Environmental and Legal
PMA	Plan for Modernization of Agriculture
PARI	Public Agricultural Research Institute
NARS	National Agricultural Research System
NBBB	National Biotechnology and Biosafety Bill
SR	Strategic Result
SWOT	Strength, Weakness, Opportunity and Threat
UIRI	Uganda Industrial Research Institute
CoE	Centres of Excellence

EXECUTIVE SUMMARY

The Government of Uganda (GoU) recognized the role of the agricultural sector in poverty eradication through its Vision 2040 and National Development Plan III as a key national development agenda. The aim is to "transform Ugandan society from a peasant dominated to a modern and prosperous middle-income country within 30 years". This can be achieved through increased agricultural production/productivity and agro-processing. Oil palm production is currently one of the promising ventures to achieving the vision 2040 since its commercialization in the early 2000. Oil palm commercialization has been developed through VODP I, II, and NOPP, to reduce on the vegetable oils import burden of 65% for majorly edible oil, detergents and soap production. Oil palm production in Uganda is implemented through public-private-partnership. However, there has been no clear long-term direction on oil palm research given the anticipated production increase. Therefore, this strategic plan is to provide tactical and operational guidance to research, production and policy for improved delivery of oil palm research services and products. The major strategies identified include; Generation of TIMPS to enhance production and productivity for accelerated development of the oil palm industry, enhanced stakeholder linkages, access and transfer of oil palm TIMPS, increased products and services suited for the oil palm industry, enhanced capacity for effective and sustainable oil palm research, institutional arrangements for implementation of the oil palm research, strategic plan and funding and risk management.

The strategies will be implemented through various approaches tailored towards the needs of the different stakeholders. The overall emphasis will be on increased productivity together with extensive cultivation. MAAIF will take the lead in coordination, monitoring and evaluation of efforts in the implementation of the OPRSP.

CHAPTER 1 – BACKGROUND

1.1 INTRODUCTION

1.1.1 Vision 2040 and NDP

The key guiding economic growth and development policy for Uganda is Vision 2040, which aims to "transform Ugandan society from a peasant dominated to a modern and prosperous middle-income country within 30 years". Vision 2040 is implemented through five-year serialized National Development Plans (NDPs) notable of which have been NDP I, II and now III.

Under the current NDP III (2020/21–2024/25), the national goal is to increase average household incomes and improve the quality of life of Ugandans by undertaking five objectives. These include; (i) Enhancing value addition in key growth opportunities, (ii) Strengthening the private sector to create jobs, (iii) Consolidating and increasing the stock and quality of productive infrastructure, (iv) Enhancing the productivity and social wellbeing of the population and (v) Strengthening the role of the state in guiding and facilitating development. These objectives are hinged to the aspirations of the Sustainable Development Goals (SDGs) which aim at ending poverty, protecting the planet and ensuring quality life for all by 2030.

Further, these objectives are envisaged to individually and collectively streamline investments towards increased agricultural production/productivity and agro-processing. In this regard, the NDP recognizes that agro-processing occupies a very important place in the agricultural value chain, creating backward and forward linkages between the farm and the market. Hence, for the sector to increase production, productivity and value addition for increased food security, household incomes, and export earnings, investments will aim to; increase agricultural production and productivity; improve post-harvest handling and storage, agro-processing and value addition; increase agro-processing of the selected products; increase market access and competitiveness of agricultural products in domestic and international markets; increase the mobilization, access and utilization of agricultural finance; and, strengthen institutional capacities for the delivery of agro-industrialization.

1.1.2 Agriculture Sector Strategic Plan

The agro-industrialization program of the NDP III on which the Oil Palm Strategic Plan is hinged notes that the agricultural sector will pursue the agro-industrialization agenda for inclusive employment, increased household incomes, food security and agricultural export earnings through five broad objectives, namely:

a) Strengthening production systems for sustainable production and productivity.
b) Improving agro-processing, value addition and storage.
c) Increasing agricultural product market competitiveness through improved quality and standards.
d) Increasing access to agricultural finance and insurance.
e) Strengthening the institutional, legal, policy, planning and regulatory framework for improved service delivery

1.1.3 Agricultural Research and NARO

The National Agricultural Research Organization (NARO), as an agency of MAAIF, was established as a body corporate by the National Agricultural Research Act of 2005 with the mandate to coordinate and oversee all aspects of agricultural research in Uganda. NARO comprises of a Governing Council, a Secretariat and 16 Public Agricultural Research Institutes (PARIs) spread across the country. NARO is mandated to undertake research in all aspects of agriculture including crops, livestock, fisheries, forestry, agro-machinery, natural resources and socio-economics. In conducting her research mandate, NARO broadly contributes to three programs in NDP III through development of program implementation action plans. The three programs are agro-industrialization; innovation, technology development and transfer; and regional development. In order to execute its mandate, NARO developed a 10-year strategic plan (2018/19 – 2027/28).

The vision and mission of the current NARO strategic plan are: Vision- '*A competitive society supported by a dynamic agricultural research innovation system'* and Mission-*'To innovate for sustainable agricultural transformation'*. Hence, the goal of the plan is *'To increase total factor productivity and access to agricultural research products and services* *for inclusive growth'*. In view of this, in the next 10 years NARO will focus research and innovation efforts on four broad results:

- **Strategic Result 1:** Increased niche markets for communities in the agricultural sector.
- **Strategic Result 2:** Increased research products and services suited for industry.
- **Strategic Result 3:** Communities increasingly using improved agricultural technologies and innovations.
- **Strategic Result 4:** Institutional orientation to agricultural transformation.

Although oil palm is not one of NARO's strategic crops, its potential to contribute to agro-industrialization justifies efforts aimed at increasing its production and productivity. This was the cornerstone of this strategic plan.

1.2 NATIONAL CROP RESOURCES RESEARCH INSTITUTE (NACRRI)

1.2.1 History and Evolution of NaCRRI

National Crops Resources Research Institute (NaCRRI) is one of the 16 Public Agricultural Research Institutes (PARIs) and together with the secretariat constitute the National Agricultural Research Organization (NARO). NaCRRI is located at Namulonge in Wakiso district, 27 kilometers from Kampala, along Gayaza-Zirobwe road. NaCRRI was started at Namulonge as a cotton research station under the British rule in 1947. In the period 1947–1972, the station conducted cotton production research for British Empire cotton growing countries in Africa and India. The research station was handed over to Government of Uganda on January 1st, 1972. Following the handover, the mandate of the research station was expanded to include research on food crops and livestock. However, in line with the National Agricultural Research Act (NARs Act, 2005), cotton research was moved to the National Semi-Arid Resources Research Institute based in Serere while pasture and livestock research were moved to the National Livestock Resources Research Institute.

1.2.2 Mandate of NaCRRI

The mandate of NaCRRI is to generate, develop and disseminate crop technologies and knowledge appropriate to agro-ecologies in the country. The Institute has one satellite station mainly focused on citrus research based in Kiige, Kamuli district. NaCRRI is structured into four research programs, namely: (i) Root Crops Research Program conducting research on cassava, sweet potato and yams (ii) Cereals Research Program conducting research on maize and rice (iii) Legumes Research Program conducting research on beans and soybeans while (iv) Horticulture and Oil Palm Research Program conducts research on fruits, vegetables, spices and oil palm. The institute also hosts the National Invasive Species Unit and other crosscutting functional units such as Bioscience laboratories and Uganda Biosciences Information Centre (UBIC).

Oil palm was introduced in Uganda in 1910 when the *Dura* variety was brought to Entebbe botanical gardens. Improved varieties of Tenera were later introduced between 1962 and 1972 on a trial basis. In 1972, 8 *Tenera* hybrids from Lame, Ivory Coast were planted at 4 sites: Kituza coffee research station in Mukono district, Nakabango in Jinja district, Bubukwanga in Bundibugyo district and Bugala Island in Kalangala district. However, oil palm research was temporarily halted because of political and security instability at the time. In the 1990s, comprehensive yield data analysis of fresh fruit bunches at Kituza coffee research station without any soil amendment on the trial showed yields between 14 to 17 MT per hectare per year. The study demonstrated that commercial oil palm production was viable in Uganda. The success of oil palm research trials has contributed a great deal to the current and future investment in commercial oil palm production in Uganda.

Following reforms in NARO in 2005 and upgrading of Kituza satellite station to National Coffee Research Institute, the mandate for oil palm research was transferred to NaCRRI under the Horticulture Program. Under NaCRRI, oil palm research conducts on-farm research on plantations in Kalangala and Buvuma and manages early and recent adaptive trial research plots planted in various districts across the country.

1.3 DEVELOPMENT OF THE OIL PALM RESEARCH STRATEGIC PLAN

1.3.1 Justification/Rationale for Developing the Strategic Plan

NARO has developed several strategic plans covering all its institutes including NaCRRI. Against this, however, since the inception of oil palm research in Uganda, no strategic plan for this commodity has been developed. The recent commercialization of oil palm production in Uganda has come with increased demand for research services in the oil palm sector. This increased demand has made it necessary for the Horticulture and Oil Palm Program to develop a strategic plan so as to respond to the demands of the private sector and other stakeholders in the oil palm value chain.

This strategic plan was also developed to provide strategic, tactical and operational guidance at the policy, institute and program level. Hence, it's anticipated that the strategic plan will sustain and guide oil palm research irrespective of major changes in both institute and program management. Under NARO and NaCRRI guidance, one of the responsibilities of Horticulture and Oil Palm Program (HOP) is to contribute to the attainment of middle-income status of Ugandans as aspired by the Government of Uganda in Vision 2040 and NDP-III through oil palm research. Thus, the Oil Palm Research Strategic Plan (OPRSP) was developed to serve not only as a policy document that defines the program's future interventions in oil palm research and development but also as a statement to oil palm external stakeholders of the program's commitment and contribution to making future transformation of the country a reality through the oil palm sector. Lastly, by developing this strategic plan, the program is using the opportunity to improve its performance and remain relevant in the delivery of oil palm research services and products as mandated by the NARO Act.

1.3.2 Process in the Development of the Strategic Plan

The Oil Palm Research Strategic Plan was developed using a participatory and consultative approach headed by a strategic planning committee. Interactions with several players in the oil palm sub-sector including public and private sector institutions, academia and local governments resulted in the definition of strategic objectives and laying of an action plan to execute the strategy. This was followed-up with many internal meetings for further brainstorming and review of the draft plan with guidance of an external facilitator/consultant.

The strategic planning process took the following steps;
i) Understanding and clarifying the terms of reference of the Strategic Planning Select Committee and development of the strategic framework to guide the planning process.
ii) Development of a roadmap on the strategic plan development process. This was done by the Strategic Planning Select Committee.
iii) Stakeholder consultative process to generate strategic issues that became the building blocks of the strategic plan.
iv) A SWOT Analysis facilitated by internal consultants. This involved participation of all stakeholders identified for the exercise.
v) Determining the strategic direction of oil palm research including formulation of the vision, mission, core values and the strategic objectives.
vi) Compilation of the draft strategic plan by the Strategic Planning Select Committee.
vii) Consultation with stakeholders to refine the formulation of the vision, mission, core values and the strategic objectives with their input.
viii) Sharing of the draft strategic plan with NaCRRI staff.
ix) Staff meeting to compile the final document/strategic plan.

1.3.3 Structure of the Strategic Plan

The plan is presented in six chapters as follows:

- Chapter 1: Background
- Chapter 2: Situation Analysis
- Chapter 3: Strategic Direction of NaCRRI
- Chapter 4: Results Areas, Strategic Objectives and Actions
- Chapter 5: Institutional Arrangements for implementing the plan
- Chapter 6: Funding and Risk Management

CHAPTER 2 – SITUATION ANALYSIS

2.1 STATUS OF THE OIL PALM SUB-SECTOR

The Oil palm (*Elaeis guineensis*) is a tropical plant species that thrives well on high rainfall, adequate sunlight and humid conditions within the tropics. It is grown in many countries across Africa, South America, and Southeast Asia. Oil Palm is the largest source of vegetable oil, accounting for 34% of the total world vegetable oils. This production includes both palm oil at 31% and palm kernel oil at 3.4%. In 2018, the world produced 72 million tons of palm oil with Indonesia and Malaysia producing 41 million tons (57%) and 20 million tons (27%) respectively. Other producers included; Thailand, Colombia, Nigeria, Guatemala, and Ecuador. All of these countries lie along the zone of 'optimal conditions' around the equator.

Uganda's annual demand for edible oil is currently 120,000 tons, against a production capacity of 40,000 tons. Uganda imports about 65% of crude vegetable oils to meet its national demands. As such, there is growing interest by the Government of Uganda in developing the oil palm industry for import substitution. The current areas where commercial oil palm production is carried out in Uganda are Kalangala and Buvuma districts.

Data from VODP II indicates that in 2015 palm oil production was approximately 23,000 tons representing a mean growth in production of 37% from 2010. The trade value of the crude palm oil in 2015 was US$149 million. Since 1998, the Government of Uganda has invested in domestic production and processing of vegetable oils to meet the increasing national demand. The Government of Uganda's interest in the development of palm oil partly originates from the Comprehensive African Agriculture Development Programme (CAADP), and the government's Vision 2040, National Development Plans, the National Agriculture Policy (NAP), and the Agriculture Sector Strategic Plan (ASSP). CAADP, endorsed in 2003 (Maputo Decision, 2003), was formulated to stimulate reforms in the agricultural sector that would impact on socio-economic growth and sustainable development. It is Africa's policy framework for agriculture and agriculture-led development, and an integral part of the New Partnership for Africa's Development (NEPAD). Through CAADP, Africa believes that agriculture and the food industry can be the engine for growth in Africa's largely agrarian economies, with tangible and sustainable impact on improving food security and nutrition, contributing to wealth and job creation, empowering women and enabling the expansion of exports.

2.2 CURRENT AND PREVIOUS INTERVENTIONS IMPLEMENTED ON OIL PALM IN UGANDA

2.2.1 The Vegetable Oil Development Project (VODP)

The Vegetable Oil Development Project (VODP), implemented by the Ministry of Agriculture, Animal Industry and Fisheries (MAAIF), was the government's selected strategic effort to increase domestic vegetable oil production, while also addressing rural poverty by involving farmers, and improving the health of the population through increased vegetable oil intake. The VODP implementation strategy was to be delivered through a public-private partnership arrangement where the government took sole responsibility for acquiring land for oil palm development, and the private sector partner, BIDCO Uganda Ltd., committed to providing investment, resources and technology for oil palm development and value addition.

2.2.1.1 Phase I (VODP I):

The first phase of VODP started in 2002, completed on 31 December 2011 and closed on 30 June 2012. VODP negotiated a tripartite collaboration between the government, BIDCO (and its joint venture partners), and smallholder farmers, to establish plantations and processing units for production of palm oil on Bugala Island in Kalangala district. The BIDCO conglomerate set up the nucleus estate, the palm oil mill and refinery, and established Oil Palm Uganda Limited (OPUL) to manage the plantations and processing units. The first 10,000 ha were developed on Bugala Island, with a nucleus estate of 6500 ha, plus 3500 ha of smallholder production, and the building of a mill to process 30-60 t of fresh fruit bunches per day.

2.2.1.2 Phase II (VODP II):

The second phase of the Vegetable Oil Development Project (VODP II) was approved by IFAD's Executive Board in April 2010, and by the Uganda Parliament on 29 September 2010 (IFAD, 2012). It was funded by IFAD under a loan agreement (No. 806-UG) signed on 21 October 2010, with a total project cost of US$146,175 million. Of this, US$70.38 million was from Oil Palm Uganda Limited (OPUL), US$52 million was a loan from IFAD, a GoU contribution of US$14.14 million, US$5.48 million from Kalangala Oil Palm Growers Trust (KOPGT), farmers' contribution estimated at US$3.89 million, and US$0.285 million from the Netherlands Development Organization. The project had three components; the Oil Palm Development Component, the Oil Seeds Development Component, and Project Management. Under the oil palm development component, the project partnered with BIDCO to establish 10,800 ha of oil palm in phase one and two through a nucleus estate of 6500 ha and a smallholder scheme of up to 4,300 ha. As of 2018, the nucleus estate was 6,440 ha (99% of VODP target), and smallholder plantations at 4,424 ha (94%), of which 3,020 ha had reached maturity. A VODP II supervisory report (4812-UG, November 2018) indicated that the project was on target with respect to its major outputs and outcomes. VODP II had also projected to have established a 2,500-ha smallholder oil palm scheme in Buvuma district by December 2018, and to have expanded oil palm smallholder scheme in Kalangala District to other islands, including Bunyama (207 ha) and Bubembe (119 ha). Currently, in Buvuma district, the government has purchased over 7,500 ha of land for commercial oil palm production.

2.2.2 The National Oil Palm Project (NOPP)

The National Oil Palm Project (NOPP) under the Ministry of Agriculture, Animal Industries and Fisheries (MAAIF), is designed to consolidate investments undertaken under VODP I and II to support communities producing oil palm. It is a 10-year project with the goal of 'inclusive rural transformation through oil palm investment. Now in transition, NOPP is to invest in a number of oil palm investment hubs, defined as agro-climatically suitable areas within a radius of approximately 30 km around a crude palm oil mill where at least 3,000 ha of oil palm production can be assured. Three hubs have been identified, Buvuma Island, Mayuge, and Masaka/Rakai.. The development objective of NOPP is to 'Sustainably increase rural incomes through opportunities generated by the establishment of an efficient oil palm industry that complies with modern environmental and social standards. NOPP proposes to empower communities to seize the emerging economic opportunities by developing both non-oil palm farming and non-farming livelihood activities, and to further mitigate potentially negative effects of oil palm investments in areas such as land tenure security, food security, environment and management of natural resources, and HIV/AIDS.

2.3 ACHIEVEMENTS OF OIL PALM RESEARCH IN UGANDA

2.3.1 Technology Generation and Dissemination

Research and Technology development are very critical because they enhance oil palm production and productivity. To achieve this, activities under research and technology development have been guided by the following objectives:

i) Develop improved soil fertility and oil palm tree management technologies.
ii) Identify new potential areas suitable for oil palm production in Uganda.
iii) Conduct surveillance and develop integrated management strategies for diseases threatening oil palm production in Uganda.
iv) Conduct surveillance and develop integrated pest management strategies for pests damaging oil palm in Uganda.
v) Develop and disseminate information materials through various pathways.

In an effort to achieve those objectives, research activities such as establishing adaptive trials, pest and disease surveillances, best agronomic studies, and physiology related experiments are continuously conducted to guide decision making in oil palm production.

Adaptive research trials: Between 2008 and 2018, oil palm research adaptive trials were conducted in Buvuma, Bugiri, Kibaale and Masaka with the aim of identifying new suitable areas for oil palm production. These studies revealed that yield of 17MT/Ha/Year was attainable in the study regions. With the same adaptive research initiative, more areas such as Buvuma, Mayuge and Masaka have been identified for oil palm production.

Furthermore, 29 acres of adaptive trials have established between 2016 and 2022 in Gulu (1 acre), Omoro (1 acre), Amuru (1 acre), Namulonge (3 acres), Dokolo (4 acres), Apac (3 acres), Nwoya (3 acres), Arua (3 acres), Adjumani (3 acres), Moyo (3 acres) and Zombo (3 acres) to validate suitability of oil palm production in northern Uganda.

Comprehensive yield studies: The decision to invest in oil palm sector is often guided by studies which provide evidence that the crop can profitably be grown in an intended area. The comprehensive yield studies conducted by research in Kituza, Bugala, Nakabango, Mayuge and Bundibugyo from 1971 to 2005 provided evidence that 17 ton/ha/year of oil palm can be attained in Uganda. The research findings were the basis for investment in commercial oil palm production in Kalangala.

Pest and diseases identification: In the period July 2015 to June 2017, oil palm disease and pest surveillance studies were conducted on-farm in Kibaale, Buvuma, Masaka, and Bugiri and on farmers' fields in Kalangala. These studies identified *Pestaloptiopsis*, Anthracnose, leaf blight and bunch rot as the most common diseases in the on-farm trials and in farmer fields. Other major diseases identified with low incidences included sudden wilt, bunch stalk rot and fruit rot. Based on pest surveillances conducted in 2016 and 2017, *Ryncophorous phoenicis* (F.) was found as a major oil palm pest that pauses a great challenge to oil palm production in the Lake Victoria crescent of Uganda. In 2018 and later 2019, Fusarium wilt of oil palm and Ganoderma trunk rot were identified in Kalangala respectively. These are the most important disease of oil palm in the world and with the potential to severely disrupt oil palm production in Uganda.

Collaboration: New research collaborations were initiated with Ghana Oil Palm Research Institute (OPRI) and a memorandum of understanding to guide this collaboration drafted. In addition, joint research activities and trainings were initiated between Ghana Oil Palm Research Institute and the newly formed oil palm research unit of Uganda. Similarly, partnerships have been made with CIRAD-Benin in regard to improving the genetic base of oil palm materials in Uganda especially against devastating diseases such as Fusarium wilt of oil palm.

2.3.2 Institutional Development

Capacity building: To conduct effective research, human resource is critical. Through project funding; three scientists were recruited to conduct Oil palm research. Being a recent crop in Uganda, with scanty information, the recruited staff were trained in sustainable oil palm production at the Oil palm research institute in Ghana. Similarly, one institutional research collaboration was formalized with Oil Palm Research Institute (OPRI) of Ghana to enable partnership and collaboration with institutions that have long term experience in oil palm research. In order to consolidate the research achievements realized in the past and enable their continuity the proposed projections in oil palm research strategy are partly aimed at consolidating the research achievements noted above and move them to the next level.

2.4 SWOT AND PESTEL ANALYSES OF OPR

2.4.1 SWOT Analysis of OPR

An internal review of oil palm research in the country was conducted to identify strengths, weaknesses, opportunities and threats (SWOT). The review also identified the external environment of Horticulture and Oil Palm (HOP) Program in terms of opportunities and threats. The HOP will endeavor to minimize the effect of its weaknesses and threats while safeguarding the strengths and exploiting the opportunities. The strengths, weaknesses, opportunities and threats are shown in Table 1.

Table 1: Strength, weakness, opportunities and threats of oil palm research in Uganda

	Strengths	Weaknesses
Internal	1. Good leadership and management structure 2. Strong research track record 3. Good culture of multidisciplinary research and teamwork 4. Moderate research facilities & infrastructure 5. Enabling environment and donor confidence	1. Inadequate human resource and facilities in oil palm research 2. Limited capacity to respond to private sector demands of oil palm 3. Limited linkages and collaboration in oil palm research 4. Non priority research funding for oil palm in Uganda 5. None existence of oil palm policy
	Opportunities	**Threats**
External	1. Increasing demand for quality oil palm research products and services 2. Increasing willingness of national, regional and international collaborators to work jointly with research 3. Government efforts to support oil palm production in the country 4. Increased interest by private sector to support oil palm research 5. Improved services to communities where oil palm production is taking place	1. Increasing scarcity of land for plantation agriculture in the country, and increasing land conflicts and wrangles 2. Emerging pests and diseases as a result of climate change & variability 3. Negative activism on expansion of oil palm production 4. Low interest of donor communities to support oil palm research

2.4.2 PESTEL Analysis of OPR

In the development of the OPRSP, key forces and trends were identified to have positive or negative impacts on oil palm research in the ten years of its implementation. The forces and trends were Political, Economic, Social and Technological, Ecological and Legislative (PESTEL) in nature.

Table 2: The anticipated positive and negative impacts during the OPRSP implementation period

FACTORS	Positive (Opportunities)	Negative (Threats)
Political / Government	• Uganda's Vision 2040 that aims at transforming the country's economy	• Manipulative political economy drivers • Political interference in policy implementation and changes in government policies and priorities

	• Political good will - Appreciation of research contributions to oil palm sector	• Unstable geo-political climate (regional and international)
Economic	• Availability and access to domestic & regional palm oil markets • Increased industrialization of the economy • Increased quality and diversity of palm oil products	• Small, hinterland and commodity-based economy • Fluctuating national economic growth and crude oil prices • Global economic decline and changes in palm oil market forces of demand
Social	• Increase in population size and demographic changes • Increasing palm oil consumers due to changing feeding culture	• Changes in consumer preferences and cultural value • Dynamics of communicable and non-communicable diseases arising from vegetable oil consumption
Technological	• Rapid technological advancements • Increased communication and outreach channels	• Inaccessibility and cost of rapid technological advancements
Ecological	• Increased access to adaptable oil palm germplasm • Restoration of High Conservation Areas (cultural & tourism sites) • Rejuvenation of degraded & marginal lands	• Biodiversity loss due to deforestation • Emergence of pests and diseases • Climate change and variability • Pollution of environment (lakes & rivers) • Depletion of natural resources
Legal	• Research supportive laws	• Limited enforcement of government laws • Deterrent laws e.g., the Land Law

The HOP program as the implementer of OPRS will continuously monitor trends in these factors in the course of implementing the strategic plan. The necessary action will be undertaken to promote the positive factors and mitigation measures will be undertaken to reduce on the negative impacts that are likely to compromise on research output development

2.5 ANALYSIS OF OIL PALM STAKEHOLDERS

2.5.1 Categories of oil palm Stakeholders

Stakeholder analysis was conducted purposely to identify the interests, roles/responsibilities, comparative advantages and contribution of the various stakeholders in the development and implementation of oil palm Strategic Plan. The analysis involved enlisting of the major stakeholders with complementary roles or synergies to oil palm research and the entire agricultural sector. Based on how stakeholders might influence the implementation process, four categories were identified, namely, those to keep informed, those to keep actively engaged, those to keep satisfied, and those to keep monitoring as highlighted below.

The strategic plan takes cognizance of these stakeholders and their importance, influence and impact on its activities. Hence, HOP will accordingly engage them for effective implementation of the Strategic Plan.

Table 3: Categories of stakeholders in the implementation OPRS

		INFLUENCE	
		LOW	HIGH
IMPORTANCE	HIGH	**Those to keep informed** • Individual farmers and farmer cooperatives • Research Institutions • Universities and schools • NEMA and NFA • Financial Institutions • Security Agents • Agro-input dealers • Manufacturers • Processors, marketers, Traders	**Those to keep actively engaged** • District Local governments • Private Sectors (OPUL) • MAAIF, NAADS, OWC • MoFPED, MoSTI, NPA • NARO governing Council, NAROSEC, PARIs • Donors / IFAD • Communication Agents • NGOs/environmentalists, Faith-based Institutions, CSOs • Cultural Institutions/ Kingdoms

		• Seed companies • Private sector foundation Uganda (PSFU) • Youth associations and women groups • Grain councils	• Oil Palm Value Chain actors
	LOW	**Those to keep monitoring** • Consultants • CGIAR centers, ASARECA, FARA, ICIPE, UIRI, UIA, URA, MoTWA, • Universities and other tertiary institutions	**Those to keep satisfied** • Regional bodies (EAC, AU, COMESA) • Political actors • Cultural leaders • Environmentalists • Finance Institutions, Insurance companies • WFP, FAO

2.5.2 Challenges expressed by oil palm Stakeholders

The oil palm stakeholders identified a number of challenges as follows;

- The lack of platform for information sharing
- Non-implementation, monitoring and enforcement of some policies while other policies are demotivating (e.g. land) or government position on some policies keep changing
- Lack of key policies.
- Contradictions, political interference, and competition for government resources which affect their collaboration with NaCRRI

Table 4: Challenges faced by various Stakeholders

LEVEL		WHO	INTERESTS	CHALLENGES
Primary	Direct beneficiaries / consumers of research products	Oil palm growers, processors and academia, research	Knowledge; TIMPs	Unmet demand for oil palm seedlings Inadequate funding
Secondary	Indirect beneficiaries /	Extension workers, NGOs, CBOs and media	Ethics, environmental concerns, socio-economic concerns,	May distribute un-verified, may give negative publicity that influence on funders/ donors, negative campaigns

	Change Agents		transparency and accountability	among consumers about products and environmental impact
Tertiary	Policy / decision makers, that affect research	Political leaders, MDAs, donors	Research funding and resource utilization, care about economic growth, accountability, socio-economic concerns, impact on environment	Can create negative political environment e.g. policy on GMOs, Limited/low prioritization for Oil Research funding, donors influencing research agenda that may affect research, donors may undermine local agenda e.g. import seeds

Some stakeholders were also concerned about data quality and quantity, restricted information exchange. They also noted the weak linkages and limited engagement between researchers, farmers and other key stakeholders. In this regard, weak intra- institutional linkages, protectionism and spirit of driving own agenda were highlighted. It was also noted that some stakeholders de-campaign NaCRRI technologies while others may distort NaCRRI research message. The limited capacity of NaCRRI to supply foundation and breeder seed was also reported.

2.6 CHALLENGES AND LESSONS LEARNT IN OIL PALM SUB-SECTOR

2.6.1 Challenges

- Environmental and social concerns due to deforestation and disputes over land grabbing claims in active oil palm growing districts.
- Weak oil palm seed system.
- Th long term possibility of food insecurity in some areas arising from extensive monoculture oil palm plantations.
- Negative activism about oil palm production in Uganda.
- Limited national oil palm research capacity.
- Limited land for expansion of production in some areas.

2.6.2 Lessons learned

In the course of conducting oil palm research both positive and negative lessons have been encountered and these include**;**

Positive lessons learned

a) Some areas in Ugandan have been identified to have suitable climatic conditions for oil palm production.
b) Oil palm produces the highest oil yields and income per unit area compared to other oil crops grown in Uganda.
c) Through research, knowledge and skills have been gained and acquired in oil palm production.
d) National, regional and international research institutes are available to backstop oil palm research in Uganda as long as effective partnerships and collaboration are harnessed.
e) Palm oil and oil palm bi-products can be used to generate other useful products (manure, animal feed, power, etc).
f) The crop provides opportunities for farmer institution development e.g., farmer organisations and cooperatives.
g) The crop is suitable for job creation, livelihood improvement and oil import substitution.
h) Oil palm has reduced the pressure on fishing in Lake Victoria, thus increasing on fish exports.

Negative lessons learned

a) Unpacking/rolling out of the strategic plan into midterm/annual implementation plan linked with lack of sharing of the strategic plan to all stakeholders of NARO can affect strict adherence to the implementation of the plan.
b) Weak impact assessment mechanisms and unclear feedback pathways can make implementation of the plan difficult.
c) Poor prioritisation in some of the commodities can affect eventual research activities and funding.
d) Lack of strong international partnerships affects resource mobilisation.
e) Poor publicity of NARO affects oil palm technology adoption.
f) Potential pollution in the environment especially if the fertilizer is not applied properly.
g) Overreliance on cash crop by the oil palm farmers is causing food insecurity.
h) Loss of biodiversity due to land use change from forest to oil palm production.

i) Emergence of increased land conflict in oil palm growing areas.
j) Interest of the vulnerable groups are not usually considered in research and development projects.

CHAPTER 3 – STRATEGIC DIRECTION OF NACRRI

3.1 AIM OF THE STRATEGIC PLAN

This chapter describes the strategic direction that oil palm research will take in the next 10 years. It highlights what oil palm research wants to contribute to the socio-economic development and livelihoods of stakeholders in the oil palm sector. It also specifies what it will do to achieve its vision. The chapter focuses not only on current problems and the constraints its stakeholders are facing but also on future opportunities in the sector.

3.1.1 Vision, Mission and Goal

Vision

A market responsive, client oriented and demand driven national agricultural research system.

Mission

To generate and disseminate appropriate, safe and cost-effective technologies

Goal

To enhance production, productivity and competitiveness of oil palm research products and services for inclusive growth

Table 5: Goal, key research areas and key performance indicators for the OPR

	Key performance Indicators (KPIs)
Goal: To enhance, production, productivity and competitiveness of oil palm research products	• Percentage increase in proportion of households that are food secure from (60% to 80%) • Percentage reduction in households dependent on subsistence agriculture from 68.9% to 55% • Percentage reduction in household food and nutrition security by 30%

and services for inclusive growth	• Percentage increase in marketed share of oil palm products by 20% • Percentage increase in oil palm-based business entities initiated by 25% • Percentage increase in number of jobs created in the oil palm industry by 65%. • Percentage increase in crude palm oil by 80% • Percentage reduction of amount of forex spent on importation of vegetable oil and its associated products
KRA1	• Number of TIMPs released by OPR increased by 30%
KRA2	• Percentage of number of TIMPs adopted increased by 30% • Average Farm level yields of oil palm increased by 30%
KRA3	• Marketed fresh fruit bunches increased by 95% • Number of market-oriented products generated increased by 40%
KRA4	• Funding for oil palm research increased by 50% • Number of result monitoring indicators reported on time as outlined in the M&E plan increased to 100% by 2028 • Number of complaints/grievances resolved increased to 95%

3.1.2 Core Values and Principles

Oil palm research will implement this strategic plan in cognizance of the NARO core values which are:

Inclusivity: Inclusivity creates opportunities and conducive environment for all interested stakeholders to participate in research service delivery. Implementation of this strategic plan will be highly inclusive by providing opportunities and an open and conducive environment for all interested stakeholders to fully participate, commit and support engagement in Oil palm research and development.

Subsidiarity: NARO now coordinates a pluralistic NARS, subsidiarity will be key to increasing stakeholder engagement and equity, and to give all constitutes a sense of belonging.

Transparency: Transparency is a vital value for enhancing mutual trust and building healthy partnerships. NARO will establish mechanisms for participatory resource allocation, peer review, information sharing and allow open discussion on resource mobilization ventures. NARO's pursuit for transparency will also ensure integrity in all aspects. This will promote professionalism and ethics in agricultural research service delivery.

Accountability: NARO will promote the principle of value-for-money in resource allocation, utilization and reporting among the various actors in agricultural research.

Excellence: NARO will endeavour to exceed expectations in everything the institute does, ensuring that all services provided comply to national and international standards of excellence and staff change and adapt quickly to new development challenges and opportunities in the frontiers of science.

The following principles will guide the operations of NaCRRI

a) Professionalism: NaCRRI will endeavour to have well-trained, professionally competent and self-confident staff, that will conduct oil palm research.
b) Ethics and Integrity: NaCRRI will carry out its activities with respect, honesty and truthfulness and take all reasonable measures to prevent wilful wrongdoing by its staff while holding each accountable for his/her actions.
c) Creativity and Innovation: NaCRRI will promote a culture of creativity, innovation, continuous learning, problem solving and independent thinking.

3.1.3 Stakeholders and Beneficiaries of this Strategic Plan

Table 6: The primary and secondary beneficiaries of this strategic plan.

Primary beneficiaries	Secondary beneficiaries
• NaCRRI staff • NaCRRI Advisory Committee, • NaCRRI Management, • NOPP coordination unit • Horticulture and Oil palm program staff	NARO Council, NAROSEC Top Management, MAAIF, MoTIC, MoWE, MoLG, NARO, NAADS, OWC, UNMA, UDC, UNBS, UEPB, UCA, MoH, UIRI, UNCST, UIA, MoEMD, MoICT &NG, NPA, MoTWA, MoFPED, MGLSD, OPM

It should be noted that the ultimate beneficiaries of the outputs and services emanating from the implementation of this strategic plan will be all those targeted to be engaged in the value chains of the NaCRRI mandated crops.

3.1.4 Alignment of Strategic Plan to NDP, ASSP and NARO

There is a strong linkage between this strategic plan, with the NDP III program on agro-industrialization. It is also aligned to ASSP, NARO and NaCRRI strategic plan in terms of vision, goal and objectives as is summarized in the table below.

Table 7: Alignment of the Strategic Plan to ASSP, NARO, NaCRRI and OPR

ASSP	NARO	NaCRRI	OPR
Goal: To transform the agriculture sector from subsistence farming to commercial agriculture	**Goal:** To increase total factor productivity and access to agricultural research products and services for inclusive growth	**Goal:** To increase productivity, access, utilization and value of crop TIMPs, research products and services for inclusive growth along the value chains	**Goal:** Increase oil palm productivity, access and utilization of research products and services for inclusive growth along the value chain of oil palm

Priority 1: Increasing Agricultural Production and Productivity	**Strategic Result 2:** Increase research products and services suited for industry	**Results Area 1**: Enhance technology, innovations, management practices (TIMPs) generation for different segments of the value chains	**Result Area 1:** Generation of oil palm TIMPS for sustainable oil palm production
Priority 2: Increasing access to critical farm inputs	**Strategic Result 3**: Communities increasingly using improved agricultural technologies and innovations	**Results Area 2**: Strengthen TIMPS transfer and adoption and research-extension interface	**Result Area 2:** Enhanced stakeholder linkages and transfer of oil palm TIMPS
Priority 3: Improving agricultural markets and value addition for the 12 prioritized commodities	**Strategic Result 1**: Increased niche markets for communities in agricultural sector	**Results Area 3:** Value addition, commercialization and market access	**Result Area 3**: Increased products and services suited for the oil palm industry
Priority 4: Institutional and Enabling Environment Strengthening	**Strategic Result 4**: Institutional orientation to agricultural transformation	**Results Area 4:** Enhance capacity for effective research	**Result Area 4:** Enhanced capacity for effective and sustainable oil palm research

The above alignment to ASSP, NARO and NaCRRI strategic plans underscores the fact that indeed:

- The plan will be contributing to the national vision of transforming Uganda from a peasant to a modern and prosperous country and to the goal of the ASSP of transforming the agriculture sector from subsistence farming to commercial agriculture of the NaCRRI mandated commodities.
- The plan in responding to agricultural challenges is compliant with other national strategies and plans.

- The plan indicates that oil palm research through NaCRRI is going to focus on aspects of research and development that it has mandate and comparative advantage in handling.

In view of the above, alignment, in the next 10 years, HOP will focus her efforts in four results areas, namely,

a. **Result Area 1**: Generation of TIMPS to enhance production and productivity for accelerated development of the oil palm industry
b. **Result Area 2:** Enhanced stakeholder linkages, access and transfer of oil palm TIMPS
c. **Result Area 3**: Increased products and services suited for the oil palm industry
d. **Result Area 4:** Enhanced capacity for effective and sustainable oil palm research

CHAPTER 4: RESULT AREAS, STRATEGIC OBJECTIVES AND ACTIONS

4.1 RESULT AREA 1: GENERATION OF TIMPS TO ENHANCE PRODUCTION AND PRODUCTIVITY FOR ACCELERATED DEVELOPMENT OF THE OIL PALM INDUSTRY

4.1.1 Strategic objective 1: Address effect of oil palm on the environment

Rationale for the objective: Currently, the most suitable commercial oil palm producing areas in Uganda have been identified within and around Lake Victoria and formerly on rain forested land. In this regard, oil palm research trials were established in Mayuge (2006), Bugiri (2008), Buvuma (2008), Kibaale (2008), Masindi (2000 & 2006), Kituza (1971, 2000), Masaka (2008), Bundibugyo (1971), Namulonge (2016) and Gulu (2016). With the objective of expanding oil palm production to other areas across the country, new oil palm adaptability trials have been planted in West Nile (Adjumani, Moyo, Arua and Zombo) and in Mid-Northern Uganda (Dokolo, Apac and Nwoya). Poor oil palm production practices and land use are associated with negative environmental consequences. Therefore, this requires environmentally sustainable research TIMPs suited to various locations across the country. Oil palm is a pest and disease prone heavy feeding perennial crop that requires intensive use of fertilizers, agro-chemicals to maintain high yields during its over 25-year productive period. When such inputs are used inappropriately, they affect the environment and biodiversity in environmentally sensitive areas like swamps, rivers and lakes. Further, oil palm production causes increase in gaseous emission to the atmosphere; increase in loss of biodiversity due to deforestation; increase in release of carbon dioxide and monoxide due to industrial processing and transport; and increase in methane from factory effluent that severely pollute the environment. This is made worse by the tendency of oil palm farmers to apply agro-chemical inputs excessively without observing buffers that are in close proximity to swamps, lakes and rivers. Oil palm research should thus engage various oil palm stakeholders in ensuring the safety of the environment at different stages of oil palm production. In developing technologies for oil palm production, environmental safety must be prioritized beforehand. In some countries, there have been efforts to restore the biodiversity by establishing biodiversity conservation sites designed to attract tourists, and this requires further investigation.

Strategic Actions

1. Develop environmentally friendly oil palm technologies and evaluate them for safety.
2. Support conservation and biodiversity with respect to oil palm production
3. Mainstream environmental and social safeguards in oil palm research and production.
4. Develop practices for restoration of biodiversity that integrate agrotourism concepts.

4.1.2 Strategic Objective 2: Develop climate smart oil palm TIMPS

Rationale for the objective: Oil palm is a new crop in Uganda. Uganda is environmentally and geographically unique compared to areas of origin of oil palm and historical commercial oil palm growing areas. Despite being a novel commercial crop in Uganda, oil palm production has faced various challenging factors including nutrient deficiencies, poor agronomic practices, biotic (pests, diseases and physiological disorders) and abiotic (moisture, heat and seasonal variations) stresses which result in poor yields and oil quality. This is further complicated by environmental sensitivity of oil palm coupled with a narrow traditional and research knowledge base relative to Ugandan agro-ecological conditions. The lengthy perennial nature and the much-favored monoculture farming system have made oil palm prone to various biotic and abiotic stresses and consequently intensive use of inputs that could cumulatively affect the balance of nature through displacement and environmental pollution. Oil palm research is thus tasked to develop climate smart, cost-effective and suitable technologies to avert loss in yields and quality of palm oil.

Strategic Actions:

1. Identify areas and oil palm varieties suitable for oil palm production in Uganda.
2. Develop integrated strategies for the management of diseases, pests and physiological disorders of oil palm.
3. Develop and recommend optimal agronomic and crop management practices for increased oil palm production and productivity.

4. Develop Integrated water and nutrient management options for oil palm based cropping systems.
5. Strengthen surveillance, forecasting and early warning for different oil palm pests and diseases.
6. Develop environmentally friendly oil palm technologies and evaluate for their safety.
7. Support conservation and biodiversity with respect to oil palm production.
8. Mainstream environmental and social safeguards in oil palm research and production.
9. Conduct environmental and social impact assessment to avert negative effect.

4.1.3 Strategic objective 3: Develop systematic breeding program to deliver improved oil palm genetics

Rationale of the objective: The demand for vegetable oil surpasses its supply which creates a deficit leading to importation of most vegetable oils consumed in Uganda. This situation is often brought about by low production and productivity of vegetable oil crops including oil palm. In Uganda, the low yields are mainly attributed to poor adaptability to local environmental conditions resulting in susceptibility to both abiotic and biotic stresses. The recent outbreaks of Fusarium wilt of oil palm in Kalangala exposed the risk of adopting resistant oil palm varieties that have been evaluated and accepted in foreign countries (recently from Indonesia, Malaysia, Ghana, Benin and Nigeria) without yield and adaptability studies in Uganda. Therefore, it's necessary to strategically plan on how to avert dependency on oil palm varieties bred and evaluated abroad without prior evaluation under local environmental conditions. This necessitates setting up mechanisms to evaluate oil palm varieties for adaptability in the short term and develop suitable oil palm varieties for Uganda in the long term. These mechanisms should help in meeting the growing demand for oil palm seedlings in new production areas. The increased opening of land to meet the ever-increasing demand of oil palm products calls for easy access to quality planting materials. Thus, generation of planting materials which are high yielding and well adapted to Ugandan and regional conditions offers an effective way to respond to this challenge. Expertise is available in some collaborating countries e.g., Oil Palm Research

Institute (Ghana) and others in Nigeria that could be used to assist build capacity in Uganda. Besides, the highly skilled, dedicated and multi-disciplinary research team that is available can effectively be used to design a breeding program that can develop competitive oil palm planting materials and conserve the available genetic resources for future use. This coupled with available high input technologies such as tissue culture and invitro conservation put the research team in position to generate planting materials through the following actions.

Strategic actions

1. Assemble and characterise local and introduced oil palm genetic resources.
2. Evaluate introduced oil palm germplasm for adaption and agronomic performance.
3. Initiate product profile-based oil palm breeding program.
4. Maintain and conserve oil palm genetic resource for crop improvement.
5. Identify and develop drought tolerant, dwarf and early maturing varieties.
6. Develop capacity for oil palm breeding to meet farmers', processors' and consumer needs.

4.1.4 Strategic objective 4: Enabling technologies and tools for improved oil palm production

Rationale for the objective: In Kalangala, the only active commercial oil palm producing district in Uganda with mature plantations, production of oil palm is heavily reliant on manual operations (weeding, pruning, harvesting, etc.) with limited use of mechanizations. Similarly, oil palm research has been limited to use of physical methods in implementation of its activities including surveillance for pests and diseases, data collection and storage and monitoring of research activities. As a result, drudgery, wastage and loss of quality have been widely associated with oil palm production. The absence of research tools and a common data collection and management system for oil palm research has had an effect on feasibility of field operations and how far research can be depended on.

Strategic Action:

i) Conduct basic oil palm physiological related studies for enhanced productivity.
ii) Develop appropriate oil palm mechanization tools from production to harvesting.

4.1.5 Strategic objective 5: Develop a functional seed system for oil palm in Uganda

Rationale for the Objective: Oil palm research has continued to evaluate and identify new areas that are suitable for oil palm production. The results from these studies have been the basis for the government of Uganda to roll out commercial oil palm production across the country. However, the growing demand for seed in Uganda is only reliant on imported seedlings raised by the private sector. Access to good quality oil palm seedlings has thus limited the involvement of the wide private sector especially small smallholder farmers outside the government project areas. Easing access to good quality oil palm seedlings is therefore crucial in the development of the oil palm sector in Uganda and this can be achieved through having a functional seed system for oil palm in the country.

Strategic actions

1. Profile and develop descriptors for oil palm germplasm.
2. Develop standards for oil palm seed and seedling certification.
3. Develop, maintain and accredit oil palm nurseries.
4. Develop and optimize protocols for tissue culture and for in-situ conservation of oil palm.
5. Develop capacity (human and infrastructure for conservation) along the oil palm seed value chain.

4.1.6 Strategic objective 6: Integration of OP technologies in diverse farming systems

Rationale for the objective: Monoculture is the common method used in commercial oil palm production. The long production cycle of oil palm has thus made oil palm plantations limit land use opportunities for long periods of time. It is therefore significant for research to examine sustainable technologies and practices that can diversify land use within commercial oil palm plantations.

Strategic actions:

1. Develop and optimize sustainable land management in oil palm production areas.
2. Develop integrated and sustainable oil palm production packages for different farming systems.

3. Develop technologies that promote sustainable agroforestry farming systems in oil palm production.
4. Develop practices that promote restoration of biodiversity while integrating agrotourism.

4.2 RESULT AREA 2: ENHANCED STAKEHOLDER LINKAGES, ACCESS AND TRANSFER OF OIL PALM TIMPS

4.2.1 Strategic objective 1: Strengthen researcher-private sector-farmer linkages to enhance adoption pathways for oil palm TIMPS

Rationale of the objective: Linkages among major stakeholders in the oil palm sector are dismal due to inadequate engagement punctuated with limited capacity to respond to their demands. For example, oil palm production is mainly managed by the private sector whose needs are not fully addressed by research. For most commodities undertaken by research, previous technologies and innovations generated have not matched private sector needs due to meagre assessment of needs prior to conducting research. Quite often private sector is never engaged in the development of TIMPs which compromise their adoption once generated. The research team needs to strongly interface with all stakeholders along the oil palm chain especially end-line users like farmers and the private sector during development of oil palm TIMPs. It is essential to link research to the interests of the private sector and farmers especially focusing on factors that maximize market access and profitability, while minimizing costs.

Strategic Actions

1. Map the demand of oil palm TIMPs.
2. Participatory management of oil palm best management plots (BMPs) with end-users
3. Support farmers in acquisition of favorable credit facility to finance adoption of oil palm TIMPs
4. Conduct studies on appropriateness of different uptake pathways for oil palm TIMPs.

5. Strengthen engagement with oil palm farmers through new and more vibrant platforms, channel and uptake pathways including use of mobile ICT technologies to meet the information needs of youthful farmers and other stakeholders.
6. Promote public private partnership for enhancing joint oil palm TIMPs development and promotion.
7. Engagement of end-users in oil palm TIMPs development and evaluation.

4.2.2 Strategic objective 2: Increase access to oil palm TIMPs

Rationale of the objective: Oil Palm TIMPs generated through research can be rendered useless, unless accessed by end-users. Overtime a number of oil palm TIMPs have not been accessed by farmers due to inadequate engagement of end-users during their development. The oil palm research team will adopt direct and collaborative approaches of disseminating research output to its end-users. Thus, during implementation of this strategic plan, oil palm research team will build awareness of TIMPs to extension workers, farmers and other end-users by prioritizing funding for developing and disseminating targeted Information, Education and Communication (IEC) materials. The team will adopt contemporary tools and channels which will include multi-stakeholder platforms, demonstrations, district adaptive research support teams, radio and television programs, print media / newsletters / magazines, plant clinics, field visits, media, video documentaries, books and conferences/workshops for dissemination of information to the public.

Strategic actions

1. Create awareness among farmers and other end-users of oil palm TIMPs.
2. Build capacity of extension and private sector in dissemination of generated oil palm TIMPs.
3. Information materials generated and distributed to ease access to oil palm TIMPS.
4. Facilitate demand articulation and priority setting for transfer and dissemination of locally and internationally generated oil palm TIMPs.
5. Build capacity of relevant partners in various aspects of the research and development continuum, including packaging and sharing of information and knowledge on oil palm TIMPs using modern technologies.

6. Use appropriate ICT tools to help in development and dissemination of oil palm research information, education and communication (IEC) materials, including sharing of knowledge and skills on oil palm TIMPs with the end-users.
7. Promote the establishment and use of technology impact centres (TICs), adaptive research, on-farm trials and demonstrations, farmer field schools etc for dissemination of information on oil palm TIMPs.

4.2.3 Strategic Objective 3: Establish and maintain oil palm innovations research platforms (adaptive trials, demos, FFTs)

Rationale of the objective: The research team will engage intensively with actors along oil palm value chain through a transdisciplinary multi-stakeholder platform to ensure the relevance of research, as well as the feasibility and increased uptake of oil palm TIMPs, products and services. The platform will help to raise awareness, enable and facilitate trans-disciplinary research, cooperation and information sharing among stakeholders. This will help to form a participative strong ground for large-scale dissemination and sustainable uptake of oil palm TIMPs.

Strategic actions

1. Raise awareness of oil palm TIMPS.
2. Enable and facilitate information and knowledge sharing among stakeholders.
3. Holding collaborative events (e.g., farmers field days, adaptive trials, demos).
4. Instigate public-private partnerships to improve farmers' timely access to oil palm input-output markets.
5. Stimulate links to other agri-services like e-extension, insurance, credit and policy environments.
6. Enhance visibility and awareness of oil palm activities.
7. Facilitate and contribute to wider dissemination and long-term sustainable uptake of oil palm TIMPs.
8. Build ground for policy briefs around oil palm and support advocacy efforts for an enabling policy environment.
9. Find solutions to barriers to adoption of oil palm TIMPs.

10. Facilitate a value chain analysis for the different oil palm technologies and services produced.

4.3 RESULT AREA 3: INCREASED PRODUCTS AND SERVICES SUITED FOR THE OIL PALM INDUSTRY

4.3.1 Strategic Objective 1: Improve post-harvest handling, value addition and processing of oil palm

Rationale of the objective: The aim of oil palm production is to obtain products derived from the oil palm fruits. In order to get products of acceptable standards, the quality of the fruits harvested, the handling, and the processing method are important considerations. Harvesting stage in oil palm influences handling and processing. Fruit bunches harvested when half ripe yield lower in quantity and quality of oil. On the other hand, over ripened fruit bunches lose fruits to excessive dropping in the field and rotting. Rotten fruits result in accumulation of free fatty acids in the oil which are harmful to human health. Thus, collection of fruits from the field to the factory should be done within a limited period of time to avoid fruit deterioration and buildup of free fatty acids. This is often hampered by poor road networks and bad weather during peak harvesting seasons. Fruits may also be contaminated with foreign materials such as soil which compromise the quality of oil harvested. Currently, processing of fresh fruit bunches is limited to household products such as cooking oils, cosmetics and soaps. Therefore, there is great need for research to venture into other products such as biodiesel. The quality of oil produced should always be checked to ensure adherence to the standards.

Harvesting standards should be emphasized by research, extension, local leaders and farmers. This will serve to remind stakeholders especially farmers of the dangers of harvesting and selling poor quality fruits. Harvesting standards should also be reviewed in relation to the quality of the palm oil produced to make adjustments where necessary. Research and extension should take lead in creating platforms for advocating for better roads and regular maintenance. This is a critical issue during the rainy season when fruit production is higher, yet roads for transportation are poor and fruit deterioration is most probable. Weighing points where bunches are massed awaiting delivery to processing plants should be organized and regularly cleaned to avoid contamination of fruits. Research on new product development should be emphasized in order to come up with new products and increase the product range out of oil palm.

Strategic Actions:

1. Collect and analyse data on oil palm growth, yield, time from pollination to bunch maturity and maturity to fruit rotting to inform optimum harvesting conditions.
2. Develop post-harvest standards and train and sensitize farmers on observing harvesting standards and fruit collection and handling.
3. Lobby for farm road maintenance in appropriate platforms.
4. Conduct product development research to widen the product range for oil palm in Uganda.
5. Establish standards for the manufactured products for quality validation.

4.3.2 Strategic Objective 2: Increase oil product diversification and commercialization

Rationale of the objective: Palm oil has a number of bi-products and services that can potentially be exploited to enhance income of stakeholders along its value chain. The bi-products range from animal feeds, cosmetics, soaps, fertilizers/manure and bio-fuel which is often used in oil palm mills. The services can range from agro–tourism to ecological services. Presently, very few bi-products and services are fully exploited. Hence for the beneficiaries to fully utilize the resources accruing from the oil palm sector, there is need to identify all potential bi-products and services and promote their utilization and commercialization. The research team will take an initiative to identify potential bi-products and services with commercial value that can be developed and promoted in order to widen the benefits out of the sector.

Strategic actions;

1. Identify oil palm bi-products and services with potential commercial value.
2. Develop products through value addition to the raw bi-products.

3. Promote commercialization of value-added products in local, regional and international Markets.

4.3.3 Strategic Objective 3: Increase uptake of oil palm products to national, regional and international markets

Rationale for the objective: The critical requirement for successful marketing and exportation of oil palm products are quality, quantity and reliability. Quantity requires sufficient land which unfortunately in Uganda, less of land is devoted to oil palm production. In order to overcome this challenge more commercial and smallholder farmers have to be brought on board under the nucleus model which has already been tested in Kalangala and found to be effective. The research team needs to support the nucleus model system by identifying more new areas suitable for oil palm production in the country. Furthermore, research needs focus on strategic local regional and international market studies in relation to the mode of production, bulking, value addition and marketing of oil palm and its bi-products.

Strategic actions:

1. Conduct adaptive trials to identify new areas suitable for oil palm production.
2. Develop ICT-based market information system to facilitate access to local, regional and international oil palm markets.
3. Develop mechanisms and technologies to improve farmer access to market (supply and sales).

4.4 RESULT AREA 4: ENHANCED CAPACITY FOR EFFECTIVE AND SUSTAINABLE OIL PALM RESEARCH

4.4.1 Strategic Objective 1: Strengthen governance, leadership and management of oil palm research

Rationale for the Objective: Governance, management and leadership are critical in guiding the implementation of the oil palm strategic plan. In governance, there is a National Oil Palm Project (NOPP) steering committee to provide strategic direction for the industry. The oil palm research will ensure that authority, relevant structures, processes, systems and relations within the oil palm research through which (a) decisions are made and implemented (b) roles and responsibilities are defined (c) principles, values, rules and regulations are set and implemented to achieve the mandate of the oil palm research are strengthened and upheld for proper steering, control and direction of oil palm research. The capacity of the oil palm research group to steer research will be improved, for example through revamping and revitalizing policy engagements, coordination and management structures for oil palm research. Deliberate efforts geared towards removing structural barriers that affect work in oil palm research will be enforced so that a conducive environment is provided.

Regarding leadership, oil palm research will mobilize, motivate and give guidance and direction to its staff to envision and realize the goals of the research team. Further, oil palm research will use a combination of leadership styles such as democratic, transformational, transitional and innovative leadership styles to achieve its goals. Therefore, oil palm research will support greater involvement of leaders as champions and role models in the different sections of the research group.

The oil palm research group will target management efforts in (a) better organizing and strategizing research and development activities of the institute (b) organizing and managing research, administration and volunteers (c) improving efficient and effective planning, mobilization and management of the financial and non-financial resources of the research group (d) monitoring and evaluating the research group and its activities which will professionalize management in oil palm research.

Oil palm research is currently under the same research program with horticultural and essential oils crops which mask its significance as an important crop on its own. Given the importance of the crop, efforts will be made to establish a dedicated oil palm research and development program within NARO, at NaCRRI with support systems at the relevant ZARDIs and selected PARIs.

Strategic Actions:

1. Establish a criterion for qualifying major actors (refer to core values; excellence, subsidiary etc.).
2. Map major actors to be involved in oil palm research.
3. Clear terms of reference for all actors.
4. Implement a functional Monitoring, Evaluation and Learning tool.
5. Strengthen accountability of major actors involved in oil palm research.

4.4.2 Strategic Objective 2: Set up infrastructure for oil palm research

Rationale for the Objective: Infrastructures (laboratories, greenhouses, nurseries, office space and equipment, irrigation systems, machines, vehicles and others) are key resources required in oil palm research. These are required to enable researchers conduct effective and efficient research. Presently, some of the infrastructure and equipment exist in research but in absolute status which require immediate replacement. However, there is no information, maintenance and acquisition plan for new ones. Further, the need for new infrastructure has not been ascertained because no new infrastructure development plan is in place. Thus, it is necessary to have a plan in place and devise means of safeguarding existing infrastructure and also acquire new ones.

Furthermore, land is one of the key natural resources required for oil palm research. NaCRRI as the institute hosting oil palm research is endowed with a lot of land but on-farm trials which are conducted on farmer fields are constrained by land. Oil palm being a perennial crop with a life cycle of 25 years very few or no farmer may be willing to host such trials. There is need to lease land on farmer's fields in different agro-ecologies for on-farm trials.

Strategic Actions

1. Map and prioritize infrastructure (ICT tools, screen houses, labs, equipment) needs for OPR.
2. Develop an inventory and mechanisms for acquisition, maintenance and optimum use of infrastructure.

4.4.3 Strategic Objective 3: Mobilize financial resource for oil palm research

Rationale for the Objective: Financial resources are critical in carrying out oil palm research. Based on this premise government and developing partners, elsewhere invest in financial resources to ensure research which addresses challenges in the sector. Although, government has endeavored to mobilize resources for oil palm sector through loans, this is not sustainable because funding for oil palm research requires continuous and regular funding sources which will guarantee research continuity. Furthermore, in Uganda oil palm sector is majorly managed by private sector who should also contribute towards research funding, but this has not been achieved. Therefore, for sustainable oil palm research to prevail there is need to undertake the following actions with the aim of mobilizing financial resources.

Strategic Actions:

1. Socialize oil palm research strategic plan with key stakeholders.
2. Interest and develop co-funding mechanisms with the private sector, research institutions and academia for OPR.
3. Promote effective coordination, partnership, linkages, networking and collaboration in NaCRRI, Uganda and beyond.

4.4.4 Strategic Objective 4: Strengthen human resource capacity for oil palm research

Rationale for the Objective: Oil palm production is in its initial stages in Uganda and so is its research under Ugandan environmental and geographical conditions. Thus, there is limited experienced human resource capacity in the field of oil palm research in Uganda which is exacerbated by the fact that oil palm is an introduced commercial crop whose local traditional knowledge base is non-existent. On the other hand, channels for capacity building of human resources have been limited by absence of common international oil palm research organisations (CGIAR institutes). This has pushed for dependency on foreign donors and private sector oil palm associated organisations whose interests may not match with national oil palm research in Uganda. When research findings are from outside the country there tends to be enormous donor influence on research that may affect research and also undermine the local agenda e.g., imported seed.

It is therefore prudent to develop the capacity of national oil palm research through effective training, support and motivation of professional oil palm research staff, provision of adequate number of resource persons to carry out oil palm research activities and retaining different cadres of staff required for oil palm research as opposed to dependency on project staff. NARO should (a) engage with international stakeholders and boards, and (b) participate in conferences such as the International Palm Oil Congress and Exhibition (PIPOC), where staff can gain knowledge and exposure to the oil palm industry.

Strategic Actions

1. Map human resource needs and deploy talent for oil palm research.
2. Implement a responsive human resource development plan that includes short, medium and long-term trainings.

4.4.5 Strategic Objective 5: Management of oil palm knowledge products

Rationale for the Objective: Knowledge generation and information management are critical factors in any successful research and development program, because it is through communication that stakeholders get access to research outputs and services. Although commercial oil palm production is a recent undertaking in Uganda, over time knowledge and information on oil palm has been generated through research, and more is yet to come. Such knowledge and information have to be made available to end-users through appropriate and efficient systems. To-date, there has been no clear system in place to manage information and accumulated knowledge and learning generated through research. Hence the need to develop a system that can ensure that knowledge and information on oil palm is well managed and communicated to all stakeholders. This can be achieved by developing an appropriate information and management systems (IMS). The IMS will require that oil palm research is guided by 1) enabling policy, 2) efficient structures and technology, 3) constructive and supportive partners, 4) motivated and skilled people. For information to reach the end-users, NARO information management policy will be adopted to guide the process of oil palm information flow, existing ICT facilities will be upgraded to a level that can enable effective and timely delivery of information to stakeholder. Further, information and knowledge generated from oil palm has been scanty, not easily accessible due to inadequate training of data managers and thus strategic actions stated below.

Strategic Actions

1. Develop, align, and operationalize an oil palm information management strategy with NARO and NaCRRI's IMS.
2. Set up oil palm information and communication system based on appropriate application of ICTs, including hardware, software and network infrastructures.
3. Develop oil palm data base for capture, storage and retrieval of data /information shared by all partners and link it with NARO'S database.
4. Strengthen the capacity for documentation, production, dissemination and use of knowledge products such as oil palm grower's guide.
5. Strengthen local, regional and international knowledge networks for technical management of knowledge sharing.
6. Strengthen internal and external communication of best practices and oil palm research results.
7. Strengthen access and management to indigenous and exogenous knowledge in the local communities for integration into oil palm research and improving farming activities.

4.4.6 Strategic objective 6: Strengthen coordination, collaboration and partnership for oil palm research

Rationale for the Objective: Collaboration, partnerships and networks/co-ordinations are powerful mechanisms for supporting the changes called for through sharing expertise and strategies. Therefore, this will help the oil palm research to create necessary tangible improvement in stakeholders' livelihood, welfare and impact. There will be need to partner with many different organizations and individuals through pursuance of partnership and collaborative research with a range of appropriate partners who share similar values, aspirations and goals. Oil palm research needs to strengthen collaboration, partnership and synergies at NaCRRI and between NaCRRI and other agricultural institutions and across other public/non-public sectors and at national, regional and international levels. The collaboration, partnerships, and networks are envisaged to be powerful mechanisms for sharing expertise, strategies and resources e.g., through joint grant proposal development, joint project implementation, partnership, net-mapping as well as holding joint progress review meetings. The networks can connect oil palm research with other research groups within NaCRRI and with external partners through attachments, internships, sabbaticals as well as exchange visits.

Strategic Actions

1. Promote effective coordination, partnership, linkages, networking and collaboration in NaCRRI, Uganda and beyond.
2. Develop, institutionalize, operationalize, monitor and evaluate a proactive partnership strategy and management in oil palm and NaCRRI structures, systems and processes.
3. Strengthen collaboration and coordination structures, mechanisms and synergies in oil palm research at NaCRRI and between NaCRRI and across other public/non-public sectors and at national, regional and international levels.
4. Promote public private partnership for enhancing joint TIMPs development and promotion for the mandated commodities.

CHAPTER 5: INSTITUTIONAL ARRANGEMENTS FOR IMPLEMENTATION OF THE OIL PALM RESEARCH STRATEGIC PLAN

5.1 ROLES OF NARO, NACRRI AND PROGRAMME LEADERS IN IMPLEMENTATION OF THE STRATEGIC PLAN

5.1.1 Roles and responsibilities of NARO

- Coordination
- Funding
- Strategic partnerships
- Enabling policies

5.1.2 Roles and Responsibilities of NaCRRI Administration

The current administrative structure of NaCRRI, the host Institute of the Horticulture and Oil Palm Program, will govern and manage the implementation and review of the strategic plan through its committees discussed below.

Advisory Committee - The Institute is governed by an Advisory Committee that is composed of five members. They sit on a quarterly basis to review progress and deliberate on strategies to move the Institute forward. The day-to-day management of the institute is carried out by the Director of Research, who is also the Secretary to the Advisory Committee. There are several sub-committees set up to assist the Director of Research in managing the Institute. Among the various committees are the following:

(a) *Management committee* - The sub-committee is made up of all the heads of research programs and heads of Units at the Institute. They meet regularly to plan and review implementation of activities of the Institute. The committee also has the responsibility of monitoring and evaluation of projects that are conducted by staff.

(b) *Finance and budget committee* – This committee comprises the heads of programs, the finance officer and the administrative officer. Their role is to ensure equitable budget allocation and release of funds to the programs.

(c) *Report writing committee* - This sub-committee is composed of three eminent scientists and is responsible for compiling monthly and other reports. It analyses the reports and ensures that Institute staff and relevant stakeholders are informed about the progress towards implementing the plan.

(d) *Procurement committee* – Some eminent scientists sit on this committee together with the procurement officer. They make decisions on the various procurement requirements.

The committees will support the advisory committee in its work.

5.1.3 Roles and Responsibilities of Horticulture and oil palm Programme

Since the oil palm research is a mandate of the Horticulture and oil palm program, the Head of Program will be responsible for operationalizing this strategic plan at institute level. The Head of Program will guide the scientists in developing quarterly and annual operational plans that are aligned with the strategic plan. This will ensure that all program research activities are aimed at fulfilling the intended purpose of the strategic plan.

5.2 OPERATIONALIZATION OF THE STRATEGIC PLAN

5.2.1 Approval and Communication/Dissemination

It is important for all the key stakeholders to not only access this plan but also to be sensitized on their roles in its implementation. Hence, it will be the responsibility of the Institute leadership and Program head to ensure that the stakeholders, NARO and NaCRRI staff know and understand the content of this strategic plan. Similarly, they will also ensure that all plans, policies, procedures and systems developed are adequately communicated and shared with the relevant stakeholders and staff of the institute.

5.2.2 Mid-term Plans and Budgets

In order to provide for a thorough and in-depth appreciation of how, when, where and by who the strategic interventions in the plan will be delivered, a medium-term plan for the period 2024/25 – 2028/29 will be developed using a bottom-up consultative approach.

5.2.3 Annual Operational Plans and Budgets

Annually, an operational plan and budget for oil palm research will be made jointly by the Program leader together with a team of scientists for submission to NaCRRI management for consolidation into one annual operational plan and budget of the Institute. The oil palm annual operational plan will prioritize the activities that need to be executed in the strategic plan and stipulate the expected results out of these activities during the implementation process. The development of the oil palm operational plan will be guided by the overall output targets that will have been quantified in the M&E plan.

5.2.4 Immediate Plans and Budgets

On a quarterly basis, each scientist will develop a three-month plan for the actual activities to be conducted individually and/or jointly with other scientists or collaborating partners.

5.3 MONITORING AND EVALUATION

5.3.1 Monitoring and Evaluation Framework

The Horticulture and oil palm program together with NaCRRI, will invest in monitoring and evaluation (M&E) to strengthen oil palm research designs, improve implementation and justify allocation of limited resources. M&E will be participatory with different stakeholders committed to implementing their roles and responsibilities towards delivering results guided by the vision and mission of NaCRRI. This will help the program to demonstrate results (accountability) and learning in addition to improved efficiency and effectiveness. It is expected that knowledge will be generated through M&E.

5.3.2 Monitoring and Evaluation

Monitoring: The aim of monitoring will be to give regular feedback to the stakeholders about the progress in implementing this strategic plan for research and development. Information from monitoring will be used in decision-making for improved implementation of program's performance. Horticulture and oil palm program being under NaCRRI, the institute will monitor the implementation (inputs, activities and outputs) of this plan. Inputs which include financial, human, and material resources will be regularly monitored. It will also monitor activities - tasks personnel undertake to transform inputs to outputs. Outputs - products and services produced by program as specified in the results framework will also be monitored. NaCRRI as the implementing institute for oil palm research will further monitor results (outcomes and impacts). Outcomes - intermediate changes in the targeted population. These must be monitored to ensure that the changes are being realized or will be realized. Finally, the intended impacts - long-term, widespread improvement in society will also be monitored.

Evaluation: The evaluation will constitute Mid-term and End-term Evaluation of the strategic plan; NaCRRI Annual Review and Planning (APRM); Annual Program Review meetings for their compliance to oil palm research and development as stipulated in the strategic plan.

The information obtained from evaluation will assist in understanding whether what has been implemented is in tandem with proposed actions in the strategic plan. It will also provide an opportunity for reflection on the critical lessons learnt that can improve the implementation process. The evaluation process will be guided by five critical factors; relevance, impact, efficiency, effectiveness and sustainability.

5.3.3 Reporting

During the entire period of strategic plan implementation, the program will produce (a) Quarterly and annual reports to be consolidated into institute reports (b) NARO bi-annual scientific conference and c) mandatory annual technical reports to developing partners supporting the strategic plan implementation.

CHAPTER 6: FUNDING AND RISK MANAGEMENT

6.1 FINANCING THE IMPLEMENTATION OF THE STRATEGIC PLAN

The oil palm research strategy will be financed through several sources as highlighted below.

Recurrent and development budgets by GOU: The Government of Uganda through NARO often allocates funds to NaCRRI and therefore the Horticulture and Oil palm program to finance recurrent and development expenditures annually. The oil palm research will benefit from this type of funding. Through this model of funding, the oil palm research strategy implementation process will be partially financed.

Non-tax revenue: Based on the NTR policy the program is mandated to generate NTR which is remitted to the consolidated account through NARO. Part of the remitted funds will be requested to fund part of strategic plan implementation. The program needs to intensify and diversify income generations from services provided to private sector and other income generating opportunities along the oil palm value chain to fund the implementation of activities planned in the strategic plan.

Research grants: Currently the program is implementing a ten-year project being funded by IFAD, this will partially finance the plan. Similarly, the program boosts of scientists with capacity to write winning proposals that can attract substantial amount of funds to finance oil palm research strategy. Furthermore, building capacity of early career scientists will create a critical mass of scientist capable of attracting funds for oil palm research.

6.2 RISK MANAGEMENT

The program does not anticipate significant risks in implementing this strategic plan. The only concern is getting adequate funding to implement it successfully. However, as one of the strategies of ensuring that the plan is successfully implemented, a risk analysis matrix with mitigation measures will be developed to guide all partners and stakeholders. This will help the Institute and stakeholders to ensure effective implementation of the strategic plan.

BUDGET

Table 8: Costing of oil palm research strategic for a period of 10 years

Result Area		Y1	Y2	Y3	Y4	Y5	Y6	Y7	Y8	Y9	Y10	Total (UGX)	Total (USD $ M)	%
		000,000'												
RESULT AREA 1:	Generation of timps to enhance production and productivity for accelerated development of the oil palm industry	2238	1680	1378	1338	1370	1461	1554	1673	1581	1495	15768	3.94	37.5
RESULT AREA 2:	Enhanced stakeholder linkages, access and transfer of oil palm TIMPS	1081	891	861	801	760	810	780	780	795	845	8404	2.10	20
RESULT AREA 3:	Increased products and services suited for the oil palm industry	705	565	585	635	670	685	635	635	635	635	6385	1.59	15.2
RESULT AREA 4:	4Enhanced capacity for effective and sustainable oil palm research	1265	1245	1135	835	950	1195	1160	1250	1240	1210	11485	2.87	27.3
Total Investment		**5289**	**4381**	**3959**	**3609**	**3750**	**4151**	**4129**	**4338**	**4251**	**4185**	**42042**	**10.51**	**100**

APPENDICES

Appendix 1: Detailed costing of oil palm research strategic for a period of 10 years

Result Area		**Y1**	**Y2**	**Y3**	**Y4**	**Y5**	**Y6**	**Y7**	**Y8**	**Y9**	**Y10**	**Total (UGX)**	**Total (USD $ M)**
		000,000'											
RESULT AREA 1:	4.1 GENERATION OF TIMPS TO ENHANCE PRODUCTION AND PRODUCTIVITY FOR ACCELERATED DEVELOPMENT OF THE OIL PALM INDUSTRY	2238	1680	1378	1338	1370	1461	1554	1673	1581	1495	15768	3.942
Strategic objective 1:	4.1.1 Address effect of oil palm on the environment	200	200	180	190	222	211	221	240	246	240	2150	0.5375
Strategic action	1. Develop environmentally friendly oil palm technologies and evaluate for their safety	0	0	0	0	0	5	5	0	0	0	330	0.0825
	2. Support conservation and biodiversity with respect to oil palm production	0	0	0	0	0	0	0	0	0	0	690	0.1725
	3. Mainstream environmental and social safeguards in oil palm research and production	0	0	0	0	0	0	0	0	0	0	400	0.1
	4. Develop practices for restoration of biodiversity that integrate agrotourism concepts	60	60	60	60	72	66	76	90	96	90	730	0.1825

Strategic Objective 2:	4.1.2 Develop climate smart oil palm TIMPS	488	450	420	400	410	412	425	475	465	465	4410	1.1025
Strategic actions	1. Identify areas and oil palm varieties suitable for oil palm production in Uganda	60	60	70	70	70	70	70	70	60	60	660	0.165
	2. Develop integrated strategies for the management of diseases, pests and physiological disorders of oil palm.	80	80	80	50	50	50	50	50	50	50	590	0.1475
	3. Develop and recommend optimal agronomic and crop management practices for increased oil palm production and productivity.	80	80	80	90	90	92	90	90	90	90	872	0.218
	4. Develop Integrated water and nutrient management options for oil palm based cropping systems	48	50	40	40	50	50	50	60	60	60	508	0.127
	5. Strengthen surveillance, forecasting and early warning for different oil palm pests and diseases	30	30	30	30	30	50	50	50	50	50	400	0.1
	6. Develop environmentally friendly oil palm technologies and evaluate for their safety	0	0	0	0	0	0	0	0	0	0	250	0.0625
	7. Support conservation and biodiversity with respect to oil palm production	0	0	0	0	0	0	0	0	0	0	530	0.1325
	8. Mainstream environmental and social safeguards in oil palm research and production	0	0	0	0	0	0	5	5	5	5	340	0.085

	9. Conduct environmental and social impact assessment to avert negative effect	0	0					0	0	0	0	260	0.065
Strategic objective 3:	4.1.3 Develop systematic breeding program to deliver improved oil palm genetics	700	510	290	250	240	290	260	280	310	230	3360	0.84
Strategic Actions	1. Assemble and characterise local and introduced oil palm genetic resources	40	40	40	40	30	30	30	20	20	20	310	0.0775
	2. Evaluate introduced oil palm germplasm for adaption and agronomic performance	60	60	60	40	40	40	30	30	30	30	420	0.105
	3. Initiate product profile-based oil palm breeding program	300	200	50	50	50	100	90	120	150	70	800	0.2
	4. Maintain and conserve oil palm genetic resource for crop improvement	150	100	30	30	30	30	30	30	30	30	490	0.1225
	5. Identify and develop drought tolerant, dwarf and early maturing varieties	100	60	60	40	40	40	30	30	30	30	460	0.115
	6. Develop capacity for oil palm breeding to meet farmers', processors and consumer needs	50	50	50	50	50	50	50	50	50	50	500	0.125
Strategic objective 4:	4.1.4 Enabling technologies and tools for improved oil palm production	200	140	140	150	150	170	150	170	170	170	1610	0.4025
Strategic actions	1. Conduct basic oil palm physiological related studies for enhanced productivity	100	80	80	90	90	90	90	110	110	110	950	0.2375

	2. Develop appropriate oil palm mechanization tools from production to harvesting	100	60	60	60	60	80	60	60	60	60	660	0.165
Strategic objective 5:	4.1.5 Develop a functional seed system for oil palm in Uganda	430	240	220	220	220	250	360	370	250	250	2810	0.7025
Strategic Actions	1. Profile and develop descriptors for oil palm germplasm		60	60				60	60			240	0.06
	2. Develop standards for oil palm seed and seedling certification.	40	40					50	60			190	0.0475
	3. Develop, maintain and accredit oil palm nurseries.	150		20	20	20	20	20	20	20	20	310	0.0775
	4. Develop and optimize protocols for tissue culture and for in situ conservation of oil palm	100			60	60	60	60	60	60	60	520	0.13
	5. Develop capacity (human and infrastructure for conservation) along the oil palm seed value chain	140	140	140	140	140	170	170	170	170	170	1550	0.3875
Strategic objective 6:	4.1.6 Integration of OP technologies in diverse farming systems	220	140	128	128	128	128	138	138	140	140	1428	0.357
Strategic actions	1. Develop and optimize sustainable land management in oil palm production areas.	50	15	15	15	15	25	25	25	25	25	235	0.05875
	2. Develop integrated and sustainable oil palm production packages for different farming systems	70	60	60	60	60	50	50	50	50	50	560	0.14

	3. Develop technologies that promote sustainable agroforestry farming systems in oil palm production	40	25	25	25	25	25	35	35	35	35	305	0.07625
	4. Develop practices that promote restoration of biodiversity while integrating agrotourism.	60	40	28	28	28	28	28	28	30	30	328	0.082
RESULT AREA 2:	4.2 ENHANCED STAKEHOLDER LINKAGES, ACCESS AND TRANSFER OF OIL PALM TIMPS	1081	891	861	801	760	810	780	780	795	845	8404	2.101
Strategic objective 1:	4.2.1 Strengthen researcher-private sector-farmer linkages to enhance adoption pathways for oil palm TIMPS	320	200	180	120	115	165	200	220	190	190	1900	0.475
Strategic actions	1. Map the demand of oil palm TIMPs	35	30	10	0	0	0	0	30	30	30	165	0.04125
	2. Participatory management of oil palm best management plots (BMPs) with end-users	70	50	50	50	50	60	60	60	60	60	570	0.1425
	3. Support farmers in acquisition of favorable credit facility to finance adoption of oil palm TIMPs	15	10	10	10	10	10	10	10	10	10	105	0.02625
	4. Conduct studies on appropriateness of different uptake pathways for oil palm TIMPs	50	50	50	0	0	40	60	50	20	20	340	0.085

	5. Strengthen engagement with oil palm farmers through new and more vibrant platforms, channel and uptake pathways including use of mobile ICT technologies to meet the information needs of youthful farmers and other stakeholders	120	30	30	30	30	30	30	30	30	30	390	0.0975
	6. Promote public private partnership for enhancing joint oil palm TIMPs development and promotion.	15	15	15	15	15	15	30	30	30	30	210	0.0525
	7. Engagement of end-users in oil palm TIMPs development and evaluation	15	15	15	15	10	10	10	10	10	10	120	0.03
Strategic objective 2:	4.2.2 Increase access to oil palm TIMPs	500	430	420	400	390	390	325	325	350	350	3880	0.97
Strategic actions	1. Create awareness among farmers and other end-users of oil palm TIMPs	80	60	60	60	60	60	60	60	60	60	620	0.155
	2. Build capacity of extension and private sector in dissemination of generated oil palm TIMPs	70	60	60	60	60	60	60	60	60	60	610	0.1525
	3. Information materials generated and distributed to ease access to oil palm TIMPS	30	20	20	20	20	20	20	20	40	40	250	0.0625

	4. Facilitate demand articulation and priority setting for transfer and dissemination of locally and internationally generated oil palm TIMPs	60	60	50	50	40	40	15	15	20	20	370	0.0925
	5. Build capacity of relevant partners in various aspects of the research and development continuum, including packaging and sharing of information and knowledge on oil palm TIMPs using modern technologies.	30	30	30	30	30	30	30	30	30	30	30	0.0075
	6. Use appropriate ICT tools to help in development and disseminate oil palm research information, education and communication (IEC) materials, including sharing of knowledge and skills on oil palm TIMPs with the end-users	100	100	100	80	80	80	80	80	80	80	860	0.215
	7. Promote the establishment and use of technology impact centres (TICs), adaptive research, on-farm trials and demonstrations, farmer field schools etc for dissemination of information on oil palm TIMPs	130	100	100	100	100	100	60	60	60	60	870	0.2175
Strategic Objective 3:	4.2.3 Establish and maintain oil palm innovations research platforms (adaptive trials, demos, FFTs)	261	261	261	281	255	255	255	235	255	305	2624	0.656

Strategic actions	1. Raise awareness of oil palm TIMPS	40	40	40	40	40	40	40	40	40	40	400	0.1
	2. Enable and facilitate information and knowledge sharing among stakeholders	50	50	50	50	30	30	30	30	50	50	420	0.105
	3. Holding collaborative events (e.g. farmers field days, adaptive trials, demos)	30	30	30	50	50	50	50	30	30	80	430	0.1075
	4. Instigate public-private partnerships to improve farmers' timely access to oil palm input-output markets	20	20	20	20	20	20	20	20	20	20	200	0.05
	5. Stimulate links to other agri-services like e-extension, insurance, credit and policy environments	20	20	20	20	20	20	20	20	20	20	200	0.05
	6. Enhance visibility and awareness of oil palm activities	20	20	20	20	35	35	35	35	35	35	290	0.0725
	7. Facilitate and contribute to wider dissemination and long-term sustainable uptake of oil palm TIMPs	20	20	20	20	20	20	20	20	20	20	200	0.05
	8. Build ground for policy briefs around oil palm and support advocacy efforts for an enabling policy environment.	10	10	10	10	10	10	10	10	10	10	100	0.025
	9. Find solutions to barriers to adoption of oil palm TIMPs	15	15	15	15	15	15	15	15	15	15	150	0.0375

	10. Facilitate a value chain analysis for the different oil palm technologies and services produced	36	36	36	36	15	15	15	15	15	15	234	0.0585
RESULT AREA 3:	4.3 INCREASED PRODUCTS AND SERVICES SUITED FOR THE OIL PALM INDUSTRY	705	565	585	635	670	685	635	635	635	635	6385	1.59625
Strategic Objective 1:	4.3.1 Improve post-harvest handling, value addition and processing of oil palm	365	285	305	355	390	390	340	340	340	340	3450	0.8625
Strategic actions	1. Collect and analyse data on oil palm growth, yield, time from pollination to bunch maturity and maturity to fruit rotting to inform optimum harvesting conditions.	140	140	140	140	150	150	150	150	150	150	1460	0.365
	2. Develop post-harvest standards and train and sensitize farmers on observing harvesting standards and fruit collection and handling	35	35	55	55	80	80	80	80	80	80	660	0.165
	3. Lobby for farm road maintenance in appropriate platforms.	30	30	30	30	30	30	30	30	30	30	300	0.075
	4. Conduct product development research to widen the product range for oil palm in Uganda	80	80	80	80	80	80	80	80	80	80	800	0.2
	5. Establish standards for the manufactured products for quality validation	80	0	0	50	50	50	0	0	0	0	230	0.0575
Strategic Objective 2:	4.3.2 Increase oil product diversification and commercialisation	170	110	110	110	110	125	125	125	125	125	1235	0.30875

Strategic actions	1. Identify oil palm bi-products and services with potential commercial value	60	40	40	40	40	40	40	40	40	40	420	0.105
	2. Develop products through value addition to the raw bi-products	80	40	40	40	40	40	40	40	40	40	440	0.11
	3. Promote commercialization of value-added products in local, regional and international Markets	30	30	30	30	30	45	45	45	45	45	375	0.09375
Strategic Objective 3:	4.3.3 Increase uptake of oil palm products to national, regional and international markets	170	170	170	170	170	170	170	170	170	170	1700	0.425
Strategic actions	1. Conduct adaptive trials to identify new areas suitable for oil palm production	60	60	60	60	60	60	60	60	60	60	600	0.15
	2. Develop ICT-based market information system to facilitate access to local, regional and international oil palm markets	40	40	40	40	40	40	40	40	40	40	400	0.1
	3. Develop mechanisms and technologies to improve farmer access to market (supply and sales).	70	70	70	70	70	70	70	70	70	70	700	0.175
RESULT AREA 4:	4.4 ENHANCED CAPACITY FOR EFFECTIVE AND SUSTAINABLE OIL PALM RESEARCH	1265	1245	1135	835	950	1195	1160	1250	1240	1210	11485	2.87125
Strategic Objective1:	4.4.1 Strengthen governance, leadership and management of oil palm research	150	130	100	100	115	135	100	100	130	100	1160	0.29

Strategic actions	1. Establish a criterion for qualifying major actors (refer to core values; excellence, subsidiary etc.)	20	20	20	20	20	20	20	20	20	20	200	0.05
	2. Map major actors to be involved in oil palm research	30	30			15	15			30		120	0.03
	3. Clear terms of reference for all actors.	20					20					40	0.01
	4. Implement a functional Monitoring, Evaluation and Learning tool.	60	60	60	60	60	60	60	60	60	60	600	0.15
	5. Strengthen accountability of major actors involved in oil palm research	20	20	20	20	20	20	20	20	20	20	200	0.05
Strategic Objective 2:	4.4.2 Set up infrastructure for oil palm research	170	170	170	0	100	250	250	260	260	260	1890	0.4725
Strategic actions	1. Map and prioritize infrastructure (ICT tools, screen houses, labs, equipment) needs for OPR	150	150	150	0	50	200	200	200	200	200	1500	0.375
	2. Develop an inventory and mechanisms for acquisition, maintenance and optimum use of infrastructure	20	20	20	0	50	50	50	60	60	60	390	0.0975
Strategic Objective 3:	4.4.3 Mobilize financial resource for oil palm research	120	120	120	60	60	100	100	100	60	60	900	0.225
Strategic actions	1. Socialize Oil Palm Research strategic plan with key stakeholders.	60	60	60			40	40	40			300	0.075

	2. Interest and develop co-funding mechanisms with the private sector, research institutions and academia for OPR.	20	20	20	20	20	20	20	20	20	20	200	0.05
	3. Promote effective coordination, partnership, linkages, networking and collaboration in NaCRRI, Uganda and beyond.	40	40	40	40	40	40	40	40	40	40	400	0.1
Strategic Objective 4:	4.4.4 Strengthen human resource capacity for oil palm research	390	390	390	390	390	390	390	390	390	390	3900	0.975
Strategic actions	1. Map human resource needs and deploy talent for oil palm research.	300	300	300	300	300	300	300	300	300	300	3000	0.75
	2. Implement a responsive human resource development plan that includes short-, medium- and long-term trainings.	90	90	90	90	90	90	90	90	90	90	900	0.225
Strategic Objective 5:	4.4.5 Management of oil palm knowledge products	360	360	280	230	230	260	260	280	280	280	2820	0.705
Strategic actions	1. Develop, align, and operationalize an oil palm information management strategy with NARO and NaCRRI's IMS	70	70	30	30	30	60	60	60	60	60	530	0.1325
	2. Set up oil palm information and communication system based on appropriate application of ICTs, including hardware, software and network infrastructures	100	100	100	50	50	50	50	50	50	50	650	0.1625

	3. Develop oil palm data base for capture, storage and retrieval of data /information shared by all partners and link it with NARO'S database	80	80	40	40	40	40	55	55	55	55	540	0.135
	4. Strengthen the capacity for documentation, production, dissemination and use of knowledge products such as oil palm grower's guide.	45	45	45	45	45	45	20	20	20	20	350	0.0875
	5. Strengthen local, regional and international knowledge networks for technical and management knowledge sharing	35	35	35	35	30	30	30	50	50	50	380	0.095
	6. Strengthen internal and external communication of best practices and oil palm research results	15	15	15	15	10	10	20	20	20	20	160	0.04
	7. Strengthen access and management to indigenous and exogenous knowledge in the local communities for integration into oil palm research and improving farming activities	15	15	15	15	25	25	25	25	25	25	210	0.0525
Strategic Objective 6:	4.6 Strengthen coordination, collaboration and partnership for oil palm research	75	75	75	55	55	60	60	120	120	120	815	0.20375
Strategic actions	1. Promote effective coordination, partnership, linkages, networking and collaboration in NaCRRI, Uganda and beyond	20	20	20	20	20	20	20	20	20	20	200	0.05

	2. Develop, institutionalize, operationalize, monitor and evaluate a proactive partnership strategy and management in oil palm and NaCRRI structures, systems and processes	20	20	20	0	0	0	0	60	60	60	240	0.06
	3. Strengthen collaboration and coordination structures, mechanisms and synergies in oil palm research at NaCRRI and between NaCRRI and across other public/non-public sectors and at national, regional and international levels	20	20	20	20	20	25	25	25	25	25	225	0.05625
	4. Promote public private partnership for enhancing joint TIMPs development and promotion for the mandated commodities	15	15	15	15	15	15	15	15	15	15	150	0.0375
Total Investment		**5289**	**4381**	**3959**	**3609**	**3750**	**4151**	**4129**	**4338**	**4251**	**4185**	**42042**	**10.5105**